FDISP

FRACTION STUDIO PROPORTIONAL TOWERS

4-Way Multi-Notational Fraction Blocks (1/1 to 1/12)

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · **4·Fractions** · 5·Primes · 6·Algebra · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

CARD 04 / 10 • FRAC-01

MATHEMATICAL REFERENCE

1. WHAT IS THIS MANIPULATIVE?

PROPORTIONAL HEIGHT & 4-WAY MULTI-NOTATION

Fractions are proportional lengths on a 7mm grid.
Each block face shows fraction, decimal, and percent:
[1/3] (Symbol) = 0.333 (Decimal) = 33.3% (Percent)
Equal heights prove mathematical equivalence!

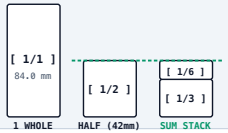
UNIVERSAL 84.00 MM WHOLE DATUM

- 1 Whole Datum: [1/1] = 12U = 84.00 mm reference rod
- Unit Base: [1/12] = 1U = 7.00 mm (Matches [2])
- Sub-Units: [1/2]=42mm, [1/3]=28mm, [1/4]=21mm
- Connectors: 1.85mm plug & recessed stacking socket

TWIN COMPARATOR WELL LID & STAGE

Drop fractional blocks into lid comparison stages;
common denominators are visible instantly!

PROPORTIONAL EQUIVALENCE: 1/3 + 1/6 = 1/2



FLUSH SUMMIT = 50%
33.3% + 16.7% = 50.0%
120° + 60° = 180°

2. KIT PARTS INVENTORY (80 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Whole [1/1]	1x	84.00 mm (12U = 1 Whole)	Classic White
Halves [1/2]	2x	42.00 mm (6U = 50.0%)	Warm Gold
Thirds [1/3]	3x	28.00 mm (4U = 33.3%)	Emerald Green
Fourths [1/4]	4x	21.00 mm (3U = 25.0%)	Royal Purple
Fifths [1/5]	5x	16.80 mm (2.4U = 20.0%)	Sky Blue
Sixths [1/6]	6x	14.00 mm (2U = 16.7%)	Coral Orange
Eighths [1/8]	8x	10.50 mm (1.5U = 12.5%)	Warm Gold
Tenths [1/10]	10x	8.40 mm (1.2U = 10.0%)	Sky Blue
Twelfths [1/12]	12x	7.00 mm (1U = 8.3%)	Slate Gray
Mini-Fractions	27x	7.00 mm Fractional Units	Assorted Family
3x1 Box & Lid	2x	49.0 mm Storage & Stage	Matte Charcoal

TOTAL: 78 Proportional Math Blocks + Storage Bin + Stage Lid = 80 Parts

FAMILY COLOR CODING SCHEME

- Color-Coded Denominators: Red=1, Orange=1/2, Green=1/3.
- Distinct Hues: Blue=1/4, Purple=1/6, Gold=1/12 unit base.
- Embossed Numerals: High-contrast white debossed numerals.
- Visual Verification: Identical colors share denominators.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Stand each fraction family in its designated graduated slot. All full families level off at the 84.00 mm rim. A missing piece leaves an immediately visible gap. Comparison stage lid snaps flush for silent zero-rattle carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF RATIONAL ARITHMETIC

Proportional height is strictly preserved under common denominator fraction addition.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 04 OF 10 • DOMAIN: PROPORTIONAL TOWERS & MULTI-NOTATION EQUIVALENCE • 84.0mm 1/1 WHOLE • 4-WAY COMPARISON FACES

LESSON GUIDE: COMMON DENOMINATOR ADDITION & EQUIVALENCE

Step-by-Step Hands-On Tutorial: 1/3 + 1/4 = 7/12 & 4-Way Multi-Notation

Grades 3–8 Fractions & Proportions • Hands-On Classroom & Home STEM Studio Series

CARD 04 / 10 • FRAC-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/fraction-studio/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.5.NF.A.1 (Unlike Denominators)
- CCSS.MATH.CONTENT.4.NF.A.2 (Fraction Comparisons)
- TEKS 111.7.b.3.H (Rational Operations)

Grades 3–8 Fractions & Proportions

MATHEMATICAL INQUIRY

Which smaller block family levels off flush with both fraction towers in the comparison slot?

1 STEP 1: SEAT 1/1 WHOLE IN REFERENCE WELL

Place the 84.00 mm Whole [1/1] block in the left reference slot.
Its 4 faces show: 1/1, 1.00, 100%, and 360°.
This physical block establishes the 100% benchmark height.

2 STEP 2: STACK 1/3 AND 1/4 BLOCKS

Place one [1/3] block (28.00 mm) in the right comparison well.
Stack one [1/4] block (21.00 mm) directly on top of the [1/3].
Total combined height: 28.00 + 21.00 = 49.00 mm.
Notice: neither thirds nor fourths alone measure this sum!

3–4 STEPS 3 & 4: TEST TWELFTHS & SOLVE

3. TEST COMMON TWELFTHS: Stack 7 twelfths: 7 × 7.00 mm = exactly 49.00 mm! Flush summit!
4. READ 4-WAY FACES: 0.333 + 0.250 = 0.583; 120° + 90° = 210°.
Calculator Verification: Press Fraction Display Icon ([FDISP] & [g][SIMP]) and run 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [+] [FDISP] → 7/12.

5 CORE PRINCIPLE: COMMON DENOMINATORS ARE COMMON SIZES

You cannot directly add thirds and fourths because their blocks are different physical heights!
Converting to twelfths (4/12 + 3/12 = 7/12) literally swaps disparate pieces for identical 7.00 mm unit blocks so their heights can be counted directly!

★ EXTENSION INVESTIGATION: ADD: 1/2 + 1/3

Seat [1/2] (42mm) and [1/3] (28mm) together. Test sixths: how many [1/6] blocks reach the exact 70.0mm summit? (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

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